COSC2129 Artificial Intelligence
Semester 3 2022
Week 2 – Search Lab

Lab Objectives

- To reinforce your understanding of basic search methods.
- To acquaint you with the operation and presentation of AI-Search software

Preparation – N-Puzzle

In Google Chrome, browse to the following address and open the **AI-Search Software (2015)**

http://www.cs.rmit.edu.au/AI-Search/

During lectures you have considered the role and application of search in Artificial Intelligence. In particular you have considered the Depth-First and Breadth-First search methods.

Lab Activities

Lab Activity 1 – Depth-First Search Algorithm

1. Set the following **Initial state** and **Goal state**

Initial state			**Goal state**		
0	2	3	1	2	3
1	4	5	8	4	5
8	7	6	0	7	6

2. In the **Search algorithm** dropdown, choose **Depth-First Search**
3. In the **Control mode** dropdown, choose **Single-step**
4. Click the **Start** button to begin the search process
5. Use **Next** and **Back** buttons to step through this problem. Watching and verifying the algorithm text coincides with the algorithm animation.
6. When the goal is reached, the solution path will be highlighted. Take notes of the length of the solution path, and the number of nodes in the open and closed list
7. Click the **Reset** button

8. Update the **Initial state** and **Goal state**

Initial state			**Goal state**		
0	2	3	1	0	3
1	4	5	8	2	5
8	7	6	7	4	6

9. Repeat Step 5 to Step 7
10. In the **Control mode** dropdown, choose **Burst mode** to see a visual overview of the search space tree constructed by a **Depth-First algorithm**
11. Begin burst mode searching through this problem by clicking the **Start** button. Take notes of the shape of the search tree, and the way in which it grows
12. **Reset** the problem and try your own **Initial state** and **Goal state** using the **Burst mode.** Again take notes of the shape of the search tree, and the way in which it grows

Lab Activity 2 – Breadth-First Search Algorithm

1. Click the **Reset** button
2. In the **Search algorithm** dropdown, choose **Breadth-Frist Search**
3. In the **Control mode** dropdown, choose **Single-step**
4. Set the following **Initial state** and **Goal state**

Initial state			**Goal state**		
0	2	3	1	2	3
1	4	5	8	4	5
8	7	6	0	7	6

5. Click the **Start** button to begin the search process
6. Use **Next** and **Back** buttons to step through this problem. Watching and verifying the algorithm text coincides with the algorithm animation.
7. Once the goal is reached, click the **Reset** button
8. In the **Control mode** dropdown, choose **Burst mode** to see a visual overview of the search space tree constructed by **Breadth-First Search**
9. Begin burst mode searching through this problem by clicking the **Start** button. Take notes of the shape of the search tree, and the way in which it grows
10. **Reset** the problem and try your own **Initial state** and **Goal state** using the **Burst mode.** Again take notes of the shape of the search tree, and the way in which it grows

Lab Activity 3 – Comparison of Depth-First Search and Breadth-First Search

1. You can compare **Depth-First Search** and **Breadth-First Search** with a Burst Mode Algorithm Race. Either team up with the person sitting next to you, or invoke a second copy of the animation page
2. Choose **Depth-First Search** for one instance and **Breadth-First Search** for the other
3. Set both instances with the following **Initial state** and **Goal state**

Initial state			**Goal state**		
0	2	3	1	2	3
1	4	5	8	0	4
8	7	6	7	6	5

4. Click **Start** simultaneously.
5. Take notes of the shape of the search tree, and the way in which it grows
6. Try a few problems of your own choice. Try to find some problems that the **Depth-First Search** can solve quickly and some others that the **Breadth-First Search** can solve quickly

Lab Activity 4 – Relative Size of Search Space Tree

1. From any given starting 8-Puzzle game board, there are a very large number of possible game boards that could be produced by following the rules of the puzzle. If we *assume* that from any possible arrangement of the 8-Puzzle we can get to any other possible arrangement of the 8-Puzzle, we essentially have a permutations problem. How many permutations of the 8-Puzzle board exist?

2. To gain some small appreciation of the relatively small portion of the "entire" search space some algorithms actually explore, have a look at the following two problems

Initial state			**Goal state**		
1	2	3	1	2	4
4	5	6	3	5	6
7	8	0	7	8	0

Initial state			**Goal state**		
1	2	3	1	2	3
4	5	6	8	0	4
7	8	0	7	6	5

The animation shows the relationship between the underlying search space and the nodes that are actually realised in computer memory.